

FnO data acquisition

Work completed and current position | Snapshot: 25 Sep 2026, 14:49 IST

Purpose

Build an auditable market-data foundation for NSE FnO stock underlyings. Strategy rules, indicators, candidates, AI results, option bid/ask history, paper P&L, alerts, Telegram, execution and trading remain outside this phase.

What we completed

- **Scope and source.** Versioned a full-universe manifest for every active FnO stock underlying. The approved source is read-only Upstox. It covers daily, 15-minute and 5-minute OHLCV; current option-contract identifiers, strike, expiry, call/put, lot and tick size; effective-dated membership; and scheduled five-minute quote snapshots. Weekly bars have a recorded Monday-to-Monday Asia/Kolkata aggregation rule from valid daily bars.
- **Controls.** Implemented raw-response and normalized storage, request/receipt/provider timestamps, delay and skew records, missing-bar and source-failure events, partial-batch checks, a hash-chained audit ledger, retention/deletion audit, a 3 GiB hard cap and a 2 GiB historical soft cap. Missing values are never filled or substituted.
- **Historical mode.** Disabled the scheduled bulk backfill. Explicit stock/interval/date requests use the approved historical API; all unrequested windows are **UNFETCHED**, not missing. Existing data is preserved. A read-only replay checked 2,436 raw records, 2,052 normalized records, two masters, 16 quote batches and 602 candle windows with no integrity or recorded-count errors at that check.
- **Reliability work.** Moved the session task to direct Python execution, fixed battery and wake settings, added bounded strict-TLS master retries, added a session-only Windows awake request, and set the task to retry every five minutes while ignoring overlapping launches. The FnO test suite passed 83 tests after v7.

Observed collection, not inferred coverage

Session	Recorded result
23 Sep	60 quote slots missed; 16 cycles recorded, one requested roughly an hour late. After-close candle retrieval crossed into the next day; 166 empty target-session responses were corrected from claimed missing bars to unavailable coverage.
24 Sep	Current-day master failed strict TLS, so all 76 quote slots were marked missed. Later diagnostic master and one RELIANCE candle check did not repair market-open coverage.
25 Sep, 14:49	Fresh master: 210 underlyings and 30,746 option contracts. Fourteen 210/210 quote cycles captured; 53 slots explicitly missed after a missed start, idle sleep and repeated process exits. The task was running at this snapshot; after-close candles were pending.

Audit position at snapshot: 7,854 ledger events verified; 62,001,142 bytes used of 3,221,225,472 bytes allowed. No five complete forward sessions have been observed, so the planned full-universe report is not yet due.

Next steps

Recommended path: collect historical data when a separately approved hypothesis needs it

The on-demand switch is a proposal in this report. The weekday forward-collection task was still enabled at the 14:49 IST snapshot.

1. Change the operating mode

Record the decision in a new manifest and audit event; disable the recurring session task and its forward-coverage heartbeat; leave the bulk backfill disabled. Preserve and verify all raw files and the ledger before any retention or move. The five-session full-universe report would be replaced by scoped reports for requested datasets.

2. Define one data request

For a separately approved hypothesis, specify the exact instruments, date range, daily/15-minute/5-minute intervals and whether point-in-time FnO membership is required. Estimate storage and request volume before fetching. Keep the 3 GiB cap, source allowlist and read-only access.

3. Fetch in capped batches

Use the Upstox historical API only for the requested windows. Store original responses, normalized bars, source and manifest hashes, request times, failures and incomplete batches. Record untouched windows as UNFETCHED. Derive weekly bars only under the versioned rule and only when required daily bars are valid.

4. Check quality before testing

Report coverage by underlying, interval and date; expected versus received bars; invalid timestamps; provider errors; partial responses; delays; storage; and deletion audit. Leave unresolved gaps visible. Review the factual report before authorizing any backtest.

Limits to decide before a hypothesis

- Historical OHLCV can be requested later; past live five-minute quote snapshots cannot be recreated. A historical API candle is not a substitute for the missed snapshot.
- Today's FnO stock list does not establish which stocks were FnO-eligible on an earlier date. A backtest requiring historical FnO membership needs a verified dated source or prospective collection; otherwise it risks selection bias.
- Current option-contract reference data does not provide historical option bid/ask or executable prices. Any option-level performance test would require a separately approved data scope.

Server decision

No always-on server is needed for scoped historical requests. If a later hypothesis needs prospective snapshots or live timing, start forward collection before the evaluation period and host its single active collector on a reliable always-on machine.

Traceability: active collection manifest v7 SHA-256 d251eb3681dd0ded46c27eff8402110d99cf26639bbf55130eea51518a7e1ff9; approved source manifest SHA-256 77d47130b0132cd4d9d66d9a284b5d32ee331db31f4079122f77e20ffbaca7c1. Snapshot evidence: C:\adipproj2\systems\proposals\ACTIVE_FNO_UNDERLYING_COLLECTION.md and the hash-chained collection ledger.